

The third row — this one *does* have a UV divergence

And its coefficient is exactly P_{gg}

$$\xi \equiv p^+/k^+, \quad \bar{\xi} \equiv 1 - \xi, \quad C_{UV}(\xi) \equiv \xi \bar{\xi} + \frac{\xi}{\bar{\xi}} + \frac{\bar{\xi}}{\xi} = \frac{P_{gg}(\xi)}{2N_c}$$

0. Verdict

Yes — there is a transverse ultraviolet divergence, and this is the first row in the series that has one. It is logarithmic, at $z \rightarrow x$, and its coefficient is

$$\frac{1}{(z-x)^2} C_{UV}(\xi) \frac{(x' - w') \cdot (y - x)}{(x' - w')^2 (y - x)^2},$$

verified to 10^{-12} on random geometries. **So this row contributes to DGLAP**, unlike the three before it — and **you must not apply the + prescription to it** (§4).

The expression itself checks out: prefactor, k^+ powers, Feynman denominator, delta function and limits all reproduce from the definitions (§1).

1. The expression is right

Assembling $\mathfrak{A}^\dagger(k^+, -w')$, $\mathfrak{C}^\dagger(k^+, w - z; p^+, z - x)$ and $\mathfrak{B}_2(k^+ - p^+, -z; p^+, -x)$:

item	from the definitions	quoted
phases	$(-i)(-1)(+i) = -1$, real	real ✓
numeric	$\frac{2}{(2\pi)^3} \cdot \frac{1}{\sqrt{2\pi}} \cdot \frac{1}{2\sqrt{2\pi}} \cdot \frac{1}{4\pi^2} = \frac{1}{64\pi^7}$	$\frac{1}{(2\pi)^3} \frac{1}{8\pi^4} = \frac{1}{64\pi^7}$ ✓
k^+	$\frac{1}{\sqrt{k^+}} \cdot \frac{\sqrt{p^+ \bar{p}^+}}{k^+ \sqrt{k^+}} \cdot \frac{\sqrt{p^+ \bar{p}^+}}{k^+} = \frac{p^+(k^+ - p^+)}{k^{+3}}$	$\frac{p^+(k^+ - p^+)}{k^{+3}}$ ✓
$\mathfrak{B}_2$ denominator	$P(y-x)^2 + K(y-z)^2$ with $P = p^+$, $K = k^+ - p^+$	same ✓
$\mathfrak{B}_2$'s $(P+K)$	$p^+ + (k^+ - p^+) = k^+$	absorbed into k^{+3} ✓
$\mathfrak{C}$'s delta	$\delta^{(2)}(w - z + \frac{p^+}{k^+}(z - x))$	same — <i>no Jacobian needed</i> , it is left in that form ✓
$\mathfrak{C}$'s kernel	$\frac{(z-x)^m}{(z-x)^2}$	same ✓
limits	$\Theta(k^+ - p^+ - \Lambda)$	$\int_\Lambda^{k^+ - \Lambda}$ ✓

(The residual overall -1 is carried by the explicit minus signs inside the two brackets, which became $[U^{ab'}(x') - U^{ab'}(w')]$ and the $ffUU - ffU$ colour structure.)

The decisive structural difference. Here $\mathfrak{C}$ is used with *first* argument k^+ and second p^+ , so its index bracket keeps **both** $\frac{k^+}{p^+}$ and $\frac{k^+}{k^+ - p^+}$. In the earlier rows the first argument was $p^+ + k^+$, which turned the second into the finite $\frac{p^+ + k^+}{k^+}$ and killed the collinear endpoint. Consistently, the limits here are the symmetric $\int_\Lambda^{k^+ - \Lambda}$ — both endpoints cut — rather than $\int_\Lambda^{\sqrt{-k^+}}$.

2. The ultraviolet divergence

Where it comes from. The new $\mathfrak{B}_2$ is the gluon vertex: it carries $\frac{1}{(x-z)^2}$, $\frac{1}{(x-w)^2}$ and $\frac{1}{(z-w)^2}$ — *squared* denominators, on separations that $\mathfrak{C}$'s $\frac{(z-x)^m}{(z-x)^2}$ already acts on. Contracting, for instance,

$$\frac{(z-x)^m}{(z-x)^2} \cdot \frac{(y-x)^i (x-z)^m}{(x-z)^2} = -\frac{(y-x)^i}{(z-x)^2},$$

a true $1/\rho^2$. The earlier rows had only single kernels and could never build one.

Measured $\rho^2 \langle \text{integrand} \rangle_\theta$ at $\rho = |z - x|$, with the six bracket structures shown separately:

rho	rho^2<F>	t1 d_perp	t2 1/xib^2	t3 1/xi	t4 1/xi^2	t5 1/xib	t6 1/(xi xib)
1e-01	+0.53239664	+0.0462	+0.1096	-0.1316	+0.3760	-0.0713	+0.2037
1e-02	+0.53241211	+0.0462	+0.1093	-0.1319	+0.3769	-0.0710	+0.2030
1e-03	+0.53241228	+0.0462	+0.1093	-0.1319	+0.3769	-0.0710	+0.2030
1e-04	+0.53241228	+0.0462	+0.1093	-0.1319	+0.3769	-0.0710	+0.2030
1e-05	+0.53241228	+0.0462	+0.1093	-0.1319	+0.3769	-0.0710	+0.2030

A nonzero constant, stable to eight digits over four decades: the integrand really is $O(1/\rho^2)$ and $\int \rho d\rho \rho^{-2}$ is logarithmically divergent. No cancellation among the six structures — each contributes.

3. The coefficient is exactly P_{gg}

The bracket carries the ξ –structures $1, \frac{1}{\xi^2}, \frac{1}{\xi}, \frac{1}{\xi^2}, \frac{1}{\xi}, \frac{1}{\xi\xi}$, and the prefactor supplies an overall $\xi\bar{\xi}$. Multiplying:

$$\xi\bar{\xi}, \quad \frac{\xi}{\bar{\xi}}, \quad \bar{\xi}, \quad \frac{\bar{\xi}}{\xi}, \quad \xi, \quad 1$$

— precisely the ingredients of $C_{UV} = \xi\bar{\xi} + \frac{\xi}{\bar{\xi}} + \frac{\bar{\xi}}{\xi} = \xi\bar{\xi} + \frac{1}{\xi\bar{\xi}} - 2$, the leftovers $\xi + \bar{\xi} + 1 = 2$ supplying the -2 . And they assemble into exactly that combination:

ξ	measured	$C_{UV}(\xi)$	ratio
10^{-4}	2029.5000	9999.0002	0.202970
0.10	1.86755226	9.201111	0.202970
0.35	0.53241228	2.623104	0.202970
0.50	0.45668317	2.250000	0.202970
0.90	1.86755226	9.201111	0.202970
$1 - 10^{-4}$	2029.5000	9999.0002	0.202970

The ratio is constant to six digits over four decades at *both* endpoints, and symmetric under $\xi \leftrightarrow \bar{\xi}$ as P_{gg} must be. The constant is the transverse structure:

$$\text{integrand} \xrightarrow{z \rightarrow x} \frac{1}{(z-x)^2} C_{UV}(\xi) \frac{(x' - w') \cdot (y - x)}{(x' - w')^2 (y - x)^2}$$

checked against the full expression to 10^{-12} on five random geometries at two values of ξ .

This is the same shape as the very first row we did, $C_{UV}(\xi) \frac{s \cdot P}{r^2}$: a product of two Weizsäcker–Williams kernels dotted together, times the splitting function, over the squared separation that is going to zero.

4. So: do not use the + prescription here

The objection is different from before, and more fundamental. In the earlier rows the $\frac{1}{p^+}$ was a *standalone soft pole*, and the obstacle to splitting was that the subtraction term was infrared divergent. Here the $\frac{1}{\xi}$ is **one half of P_{gg}** , tied by the $\xi \leftrightarrow \bar{\xi}$ symmetry to the $\frac{1}{\bar{\xi}}$ pole at the other endpoint. A + prescription applied to $\frac{1}{p^+}$ alone would tear C_{UV} apart and destroy the DGLAP structure.

What to do instead — the treatment we already built

1. **Split off the UV piece exactly**, as in the first note: $T = C_{UV}(\xi) \frac{(x' - w') \cdot (y - x)}{(x' - w')^2 (y - x)^2} \frac{1}{(z - x)^2} + T_{\text{fin}}$, with T_{fin} free of the $1/\rho^2$.
2. **Regulate the $\frac{1}{(z-x)^2}$** , either in momentum space with dimensional regularisation and $\overline{\text{MS}}$, or — equivalently and more simply — with the position–space cutoff $|z - x| > r_0, r_0 = 2e^{-\gamma_E}/\mu$.
3. **Do the ξ integral over the symmetric range**, which is what the Θ function gives you:

$$\int_{\Lambda/k^+}^{1-\Lambda/k^+} d\xi C_{UV}(\xi) = 2\ell_k - \frac{11}{6},$$

and $(2\ell_k - \frac{11}{6}) \times 2N_c = 4N_c\ell_k - \frac{11N_c}{3} = 4N_c\ell_k - b$ at $N_f = 0$: the constant is the beta-function coefficient, i.e. **coupling renormalisation**. That is the DGLAP / running-coupling structure, and it is exactly what the earlier rows lacked.

zint/src/row3_uv.py, zint/src/row3_xi.py, zint/src/row3_pref.py — every number above, written to zint/RESULTS_row3.txt.